LAB 2

String Processing in C

Due Date: 04/23/2026

Lab Overview:

For this lab, we will be learning how to process complex strings from the console input. Typically, user I/O for operating systems CLI's takes the form of processing complex string input by the user into a shell/terminal or console. These commands are then parsed and sent to a sub-process that executes them using a wide variety of system calls or predesigned routines. In this lab, we will be going over how to take complex input from the console on a line-by-line basis and parse that into understandable tokens that can be used by the system. We will be creating a helper structure that we can use to tokenize strings for current and future projects.

Core Tasks:

1. Use the provided skeleton file and implement the provided header file functions.
2. Test your code with Valgrind for memory leaks.

Task Details:

1. Tokenize the input string. Utilize strtok_r(3): http://man7.org/linux/man-pages/man3/strtok.3.html
2. Implement string_parser.c (lab1_skeleton.c is your main program that is already implemented.) If string_parser.c is implemented correctly, your program will match the output file **exactly**.
3. Make sure no memory leak exists in your program, as this program will be used for future projects.

Provided Files:

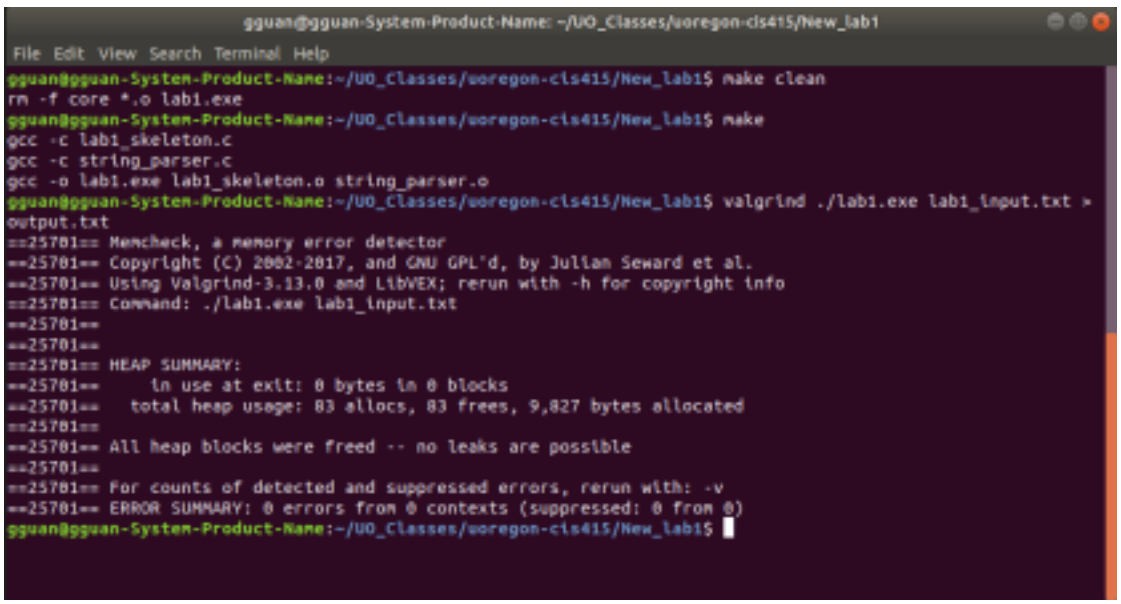
1. lab1_skeleton.c
2. string_parser.c
3. string_parser.h

4. output.txt
5. lab1_input.txt
6. makefile

Submission Requirements:

In order to receive points for this lab, the student must do the following:

1. Attend the lab and make sure your attendance is recorded.
2. Submit the zip or tar file that should contain the following.
   - string_parser.c
   - output.txt
   - a screenshot containing the following information



The naming convention of the zip/tar file while uploading to Canvas

- **UoID_duckID_LabX (an example is given below)**
  - UOID : alex
  - DuckID: 951505xxx
  - Submission for: Lab1
  - So the name of the zip/tar file should be:

    alex_951505xxx_Lab1.tar **or** alex_951505xxx_Lab1.zip